

Lung Nodule Classification on LIDC-IDRI: State of the Art and Methodological Requirements

A structured review of fourteen works (2020–2025) covering benchmark performance, evaluation validity, transformer architectures, uncertainty quantification, calibration, ordinal labelling, concept-based explainability and agentic orchestration — each mapped to the implementation tasks of this project.

PREPARED FOR	PROJECT	DATASET
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COVERS TASKS		
2, 3, 4, 5, 6		

1 Scope and method

This review supports the implementation work already under way. It is organised by the question each group of papers answers for us, not by chronology, and every entry closes with an explicit statement of how it bears on a specific task.

Papers were located through targeted searches across eight themes, then **each source was retrieved individually to confirm its title, authorship, venue, year and reported results**. Figures quoted below are those stated by the authors. Where a source was paywalled and only abstract-level detail could be confirmed, the summary is limited accordingly and says so. Claims that could not be verified against the source were excluded rather than repeated from search summaries.

Two entries predate the 2022–2026 window — references [10] and [11], from 2020. They are retained because they remain the standard points of reference for the ordinal-label problem and for calibration in medical imaging respectively, and because both bear directly on decisions we have already made.

Coverage map

PROJECT TASK	QUESTION IT RAISES	REFERENCES
Task 3 — dataset exploration	What is in LIDC-IDRI and how are labels formed?	11, 12
Task 4 — CNN baseline	What performance is credible, and what makes it credible?	1, 2, 3, 4
Task 5 — ViT comparison	Do transformers beat CNNs at this data scale?	5, 6
Task 6 — MC Dropout UQ	Which UQ method, and how is it validated?	7, 8, 9
Task 6 — calibration	How should ECE and Brier be interpreted?	9, 10
Proposal item 7 — ordinal	What to do with indeterminate nodules?	11
Proposal item 8 — explainability	Can explanations use radiologist concepts?	12
Proposal item 1 — agentic framing	Is "agentic AI" substantiated in radiology?	13, 14

Which reviewed work informs which task. Numbers refer to the reference list in section 13.

2 Executive summary

Reported accuracy on LIDC-IDRI clusters around 90–98%, but the numbers are not comparable to one another. Studies differ in how they fuse the four radiologists' malignancy ratings, whether they exclude indeterminate nodules, whether they split by patient or by nodule, and whether they evaluate on LIDC-IDRI directly or on the LUNA16 derivative. A headline figure without its protocol is close to uninterpretable.

The largest methodological risk is evaluation design, not architecture.

Tampu et al. [2] measured accuracy inflation of 5–30% from improper splitting alone. Varoquaux and Cheplygina [1] identify the same class of failure as systemic across medical imaging. This is the strongest external support for the patient-level cross-validation and the audit trail we have built into Task 04-01.

Uncertainty quantification is where the literature is thinnest and the opportunity clearest. Zahari et al. [7] is the closest published analogue to Task 6 — it compares MC Dropout, deep ensembles and their combination on 3D CT nodules — and finds the combination more reliable than MC Dropout alone. Peeters et al. [8] show uncertainty estimation improving a clinical nodule risk model. Neither couples uncertainty to an orchestrator that acts on it, which is precisely the gap this project targets.

Transformers do not have a clear win at this scale. The transformer work we could verify on lung CT relies on self-supervised pretraining over large

unlabelled cohorts [5], not on supervised fine-tuning of a few hundred labelled nodules. That is consistent with our expectation that ViT-tiny will underperform ResNet-18 on a 322-nodule dataset, and it is the honest framing for Task 5.

THE GAP THIS PROJECT CAN CLAIM

No reviewed work combines (i) a protocol-audited patient-level evaluation, (ii) decomposed epistemic/aleatoric uncertainty, and (iii) a routing policy that consumes that decomposition to decide between escalation and human referral. Individually each element is established; the composition, evaluated honestly on a stated number of nodules, is not.

3 Evaluation validity and data leakage

These three works underpin the audit structure of Task 04-01. They are the reason the notebook prints evidence of fold separation rather than asserting it.

REFERENCE 1

Machine learning for medical imaging: methodological failures and recommendations for the future

G. Varoquaux, V. Cheplygina · *npj Digital Medicine*, vol. 5, art. 48, 2022 · doi.org/10.1038/s41746-022-00592-y

A field-level critique arguing that progress in medical imaging ML is slowed less by model capacity than by how studies are designed and reported. The authors trace bias entering at every stage — dataset construction, evaluation protocol, metric selection, and the publication incentive to report a winning number — and set out recommendations for more trustworthy practice.

RELEVANCE TO OUR WORK

This is the framing document for the review comments we received on Task 4. It supports the position that AUROC 0.9005 establishes nothing on its own: what establishes it is the protocol behind it. Our six audits (A1–A6) are a direct response.

REFERENCE 2

Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images

I. E. Tampu, A. Eklund, N. Haj-Hosseini · *Scientific Data*, vol. 9, art. 580, 2022 · doi.org/10.1038/s41597-022-01618-6

A controlled measurement of what improper dataset splitting costs. Training the same models under correct and incorrect splits, the authors report performance inflated by 0.07 to 0.43 Matthews Correlation Coefficient, corresponding to 5–30% accuracy. The mechanism is that consecutive images from one volume are highly similar, so random splitting places near-duplicates on both sides of the partition.

RELEVANCE TO OUR WORK

Our dataset has exactly this structure: 75 of 187 patients contribute more than one nodule, one contributes nine, and nodules from a patient share scanner, reconstruction kernel and anatomy. This paper quantifies what we would have gained spuriously had we split on nodules — and it is the number to cite when justifying patient-level folds.

REFERENCE 3

How You Split Matters: Data Leakage and Subject Characteristics Studies in Longitudinal Brain MRI Analysis

D. J. Rumala · MICCAI Workshop on Fairness of AI in Medical Imaging (FAIMI), 2023 · arxiv.org/abs/2309.00350

Examines cross-validation splitting strategies for 3D brain MRI where subjects contribute repeated scans, and uses Grad-CAM to show the mechanism directly: under subject-level leakage, models attend to features identifying the individual rather than the pathology. The paper terms this *identity confounding*.

RELEVANCE TO OUR WORK

Explains *why* nodule-level splitting inflates scores rather than just that it does, which is the more persuasive argument in a methods section. The Grad-CAM diagnostic is also a cheap check we could adopt when Task 8's explainability work begins.

REFERENCE 4

Nodule-CLIP: Lung nodule classification based on multi-modal contrastive learning

L. Sun, M. Zhang, Y. Lu, W. Zhu, Y. Yi, F. Yan · *Computers in Biology and Medicine*, vol. 175, art. 108505, 2024 · doi.org/10.1016/j.combiomed.2024.108505

Applies contrastive multi-modal learning in the CLIP style, aligning CT image features with the structured nodule attributes recorded by radiologists, in order to learn a representation that captures the relationship between appearance, attributes and malignancy. On LIDC-IDRI the authors report a **benign/malignant classification rate of 90.6% with recall of 92.81%**.

RELEVANCE TO OUR WORK

A useful upper reference point for Task 4, and notable because it exploits the same radiologist attribute annotations (spiculation, margin, subtlety and the rest) that we cache for Task 8 but do not currently use as a training signal.

It is also published in *Computers in Biology and Medicine*, one of the two target venues named in the proposal, which makes it a good model for scope and framing.

A CAUTION ON COMPARING HEADLINE NUMBERS

Reported accuracies in this literature are not interchangeable. They depend on whether indeterminate nodules were excluded, how the four ratings were fused, the minimum number of readers required, and whether the split was by patient or nodule. Our own protocol (≥ 3 readers, median fusion, median = 3 excluded, patient-level folds, 322 nodules from 187 patients) should be stated alongside every figure we publish, and we should resist comparing our AUROC to a paper whose protocol we cannot read.

5 Transformers versus convolutional networks

REFERENCE 5

Lung Nodule-SSM: Self-Supervised Lung Nodule Detection and Classification in Thoracic CT Images

M. Noreen, F. Shaukat · arXiv preprint, 2025 · arxiv.org/abs/2505.15120

Uses a DINOv2 transformer backbone pretrained self-supervised on unlabelled CT, then fine-tuned for nodule analysis. Evaluated on LUNA16 (888 CT scans), the authors report **98.37% accuracy**.

RELEVANCE TO OUR WORK – READ CAREFULLY

Two qualifications matter before this number is used as a ViT benchmark. The task reported is nodule *detection*, not benign/malignant classification, and the dataset is LUNA16, a derivative of LIDC-IDRI with a different inclusion protocol. The figures are therefore not comparable to ours.

What it does establish is the condition under which transformers succeed here: **large-scale self-supervised pretraining on unlabelled CT**, not supervised fine-tuning of a few hundred labelled nodules. That is the honest explanation to give if ViT-tiny underperforms ResNet-18 in Task 5, and it points to a concrete future direction.

REFERENCE 6

Recent Progress in Transformer-based Medical Image Analysis

Z. Liu, Q. Lv, Z. Yang, Y. Li, C. H. Lee, L. Shen · *Computers in Biology and Medicine*, 2023 (arXiv 2022) · arxiv.org/abs/2208.06643

A survey of transformer applications across medical imaging tasks — classification, segmentation, detection, registration, synthesis and others — organised by imaging modality. Provides the taxonomy and the breadth of coverage rather than a head-to-head benchmark.

RELEVANCE TO OUR WORK

Appropriate as the general citation for the transformer arm of Task 5. We should not cite it for a CNN-versus-transformer verdict: the survey's own comparisons are drawn from heterogeneous studies, and we could not confirm a like-for-like comparison at our data scale from the abstract alone.

6 Uncertainty quantification

The three works below are the closest published context for Task 6 and should anchor its methods section.

REFERENCE 7

Uncertainty-aware image classification on 3D CT lung

R. Zahari, J. Cox, B. Obara · *Computers in Biology and Medicine*, 2024 · [sciencedirect.com/science/article/pii/S0010482524004086](https://www.sciencedirect.com/science/article/pii/S0010482524004086)

The most directly comparable study to Task 6. A three-phase framework — preprocessing and model selection, uncertainty quantification, then uncertainty measurement and *data referral* — applied to benign/malignant classification of 3D CT nodules. Three UQ methods are compared: **Monte Carlo Dropout, Deep Ensemble, and Ensemble Monte Carlo Dropout**. Eight architectures from the ResNet, DenseNet and Inception families all exceed F1 0.832, with InceptionResNetV2 highest at 0.845. The authors conclude that the **ensemble combined with MC Dropout gives more reliable uncertainty than MC Dropout alone**.

RELEVANCE TO OUR WORK

This is both our closest precedent and a warning. Their "data referral" stage is the same construct as our risk-coverage and abstention analysis, so the comparison is direct. Their finding that MC Dropout alone is the weaker estimator means Task 6 should report MC Dropout as a baseline and position deep ensembles as the natural next step, rather than presenting MC Dropout as the endpoint.

It is also published in a proposal target journal, and its existence tightens what counts as novelty for us — the differentiator has to be the orchestration and the epistemic/aleatoric decomposition driving it, not the application of MC Dropout itself.

REFERENCE 8

Enhancing a deep learning model for pulmonary nodule malignancy risk estimation in chest CT with uncertainty estimation

D. Peeters, N. Alves, K. V. Venkadesh, R. Dinnessen, Z. Saghir, E. T. Scholten, C. Schaefer-Prokop, R. Vliegenthart, M. Prokop, C. Jacobs · **European Radiology**, vol. 34, no. 10, pp. 6639–6651, 2024 · doi.org/10.1007/s00330-024-10714-7

Integrates uncertainty estimation into an existing, clinically validated deep learning algorithm for pulmonary nodule malignancy risk, and measures the effect on performance. Published in a leading clinical radiology journal by a group with an established nodule risk model. (Paywalled; summary based on the abstract and publication record.)

RELEVANCE TO OUR WORK

Evidence that uncertainty estimation is accepted as clinically meaningful in exactly our application, published in a clinical rather than a methods venue. Useful in the introduction to establish that Task 6 addresses a recognised clinical need, not only a technical one.

REFERENCE 9

Trustworthy clinical AI solutions: a unified review of uncertainty quantification in deep learning models for medical image analysis

B. Lambert, F. Forbes, A. Tucholka, S. Doyle, H. Dehaene, M. Dojat · **arXiv preprint**, 2022 · arxiv.org/abs/2210.03736

A unified review of UQ methods for medical imaging, framed around clinicians' reluctance to trust deep learning outputs. Covers the methods, the particular difficulties of medical images (dimensionality, variable quality), and — the more useful part for us — **the evaluation protocols by which uncertainty estimates should themselves be validated**, plus open problems.

RELEVANCE TO OUR WORK

The section on validating uncertainty estimates justifies the checks Task 6 already performs: whether uncertainty ranks the model's own errors, and whether a risk–coverage curve beats random referral. An uncertainty number that fails those is decorative, and this review is the citation for saying so.

REFERENCE 10

Confidence Calibration and Predictive Uncertainty Estimation for Deep Medical Image Segmentation

A. Mehrtash, W. M. Wells III, C. M. Tempany, P. Abolmaesumi, T. Kapur · **IEEE Transactions on Medical Imaging**, 2020 · arxiv.org/abs/1911.13273

Studies confidence calibration across brain, heart and prostate segmentation. Two findings transfer directly: networks trained conventionally **tend to produce overconfident predictions**, and **model ensembling improves calibration**. The authors further show that well-calibrated networks can predict segmentation quality and flag out-of-distribution inputs.

RELEVANCE TO OUR WORK – AND ITS LIMIT

This is a segmentation study, not classification, and should be cited for the calibration principle rather than as a nodule-classification result. The principle is what Task 6 measures: our ECE and Brier comparison of the deterministic baseline against the MC-averaged prediction is testing precisely the overconfidence they describe.

Their ensembling result converges with Zahari et al. [7] from a different direction, which strengthens the case for deep ensembles as the follow-on to Task 6.

REFERENCE 11**Meta Ordinal Regression Forest for Learning with Unsure Lung Nodules**

Y. Lei, H. Zhu, J. Zhang, H. Shan · **IEEE International Conference on Bioinformatics and Biomedicine (BIBM)**, 2020 · arxiv.org/abs/2012.03480

Argues that discarding indeterminate nodules — standard practice in binary work — throws away a large and clinically important part of the dataset, and reformulates the task as **ordinal regression over benign / unsure / malignant**. In the LIDC subset used, thresholding the mean rating gives **1,108 benign** (< 2.5), **1,007 unsure** ($2.5\text{--}3.5$) and **510 malignant** (> 3.5) — the discarded middle group is larger than the malignant class.

RELEVANCE TO OUR WORK

Directly addresses proposal revision item 7 (ordinal classification via CORAL/CORN) and gives a concrete precedent for treating malignancy as ordinal rather than binary.

It also bears on a decision we have already made. Our protocol excludes only median = 3, but with four readers the median can be 2.5 or 3.5, and those boundary cases are **115 of our 322 nodules (36%)** — assigned to benign and malignant respectively on the strength of a single reader. Our Task 04-01 sensitivity analysis re-scores the model without them for exactly this reason, and this paper is the citation that justifies treating it as a live methodological question rather than a detail.

9 Concept-based explainability

REFERENCE 12

Concept-based Explainable Malignancy Scoring on Pulmonary Nodules in CT Images

R. I. Dumaev, S. A. Molodyakov, L. V. Utkin · arXiv preprint, May 2024 · arxiv.org/abs/2405.17483

Combines generalised additive models with concept-based learning on LIDC-IDRI to produce an interpretable malignancy model. It detects clinically significant attributes, learns their association with the diagnostic decision, and reports each attribute's contribution, giving human-readable explanations over named concepts rather than pixel saliency. The authors report competitive classification alongside attribute scoring.

RELEVANCE TO OUR WORK

The closest published realisation of proposal item 8. Critically, it uses the same eight radiologist-assessed characteristics — subtlety, internal structure, calcification, sphericity, margin, lobulation, spiculation, texture — that our cache already stores per nodule, so the data requirement is met and Task 8 can build on this directly rather than starting from re-extraction.

Worth noting for our own analysis: concept-based studies in this area report that much of the discriminative signal is carried by nodule size, with morphological concepts contributing smaller directionally consistent effects. Our false-negative breakdown already shows the same pattern — missed lesions were smaller (median 14.2 mm against 23.6 mm) and subtler.

REFERENCE 13

Agentic AI and Large Language Models in Radiology: Opportunities and Hallucination Challenges

S. Salehi, Y. Singh, K. K. Horst, Q. A. Hathaway, B. J. Erickson · *Bioengineering (Basel)*, vol. 12, no. 12, art. 1303, 2025 · doi.org/10.3390/bioengineering12121303

Surveys agentic AI in radiology and is unusually direct about the failure modes. It reports **LLM hallucination rates of 8–15% in medical imaging contexts**, manifesting as anatomical misidentification, false findings or measurement errors. It describes role-based multi-agent orchestration — specialised agents for retrieval, analysis and quality control — as a structural mitigation, notes that **agents communicating confidence levels to one another** allows contributions to be weighted, and finds attention-based uncertainty more effective than text-stated confidence. Retrieval-augmented generation reduced hallucinations from 8% to 0% in one contrast-guidance application. The authors conclude that clinical deployment still requires human oversight and standardised evaluation.

RELEVANCE TO OUR WORK

The most important citation for the unresolved framing question in the proposal. It supports the agentic direction while documenting the hallucination risk that makes an LLM orchestrator hard to defend clinically.

The finding that agents should exchange *quantified* confidence, and that attention-based uncertainty beats self-reported confidence, is direct external support for the design of our Task 6 interface: a numeric, decomposed uncertainty contract rather than a model asserting how sure it feels.

REFERENCE 14

Conformal selective prediction with cost-aware deferral for safe clinical triage under distribution shift

Scientific Reports, 2026 · nature.com/articles/s41598-026-40637-w · also PMC13022086

Combines calibrated probabilistic modelling, conformal prediction and cost-aware deferral for clinical triage: it produces set-valued predictions with finite-sample coverage control and defers low-confidence cases to clinicians when doing so lowers an explicit expected clinical cost. *Author list not confirmed at the time of writing — verify before citing formally.*

RELEVANCE TO OUR WORK

Formalises what our abstention threshold does informally. We currently choose the threshold to hit a target sensitivity; this framework instead derives it from an explicit cost of a missed cancer versus a false alarm, which is both more defensible clinically and a clean route into the conformal prediction strand of proposal item 3.

11 Implications for our protocol

Confirmed — keep as is

Patient-level cross-validation with printed separation evidence [1][2][3]; ECE and Brier as paired calibration metrics [10]; reporting sensitivity at fixed specificity and a full false-negative breakdown rather than accuracy alone [1].

Revise — the literature is ahead of us

MC Dropout alone is the weaker estimator [7]; Task 6 should present it as a baseline with deep ensembles as the stated next step, not as the contribution. The 36% of our nodules sitting at median 2.5 or 3.5 deserve the ordinal treatment of [11] rather than a binary assignment decided by one reader.

The composition is what is unclaimed: a decomposed epistemic/aleatoric signal, validated against radiologist disagreement, feeding a routing policy that distinguishes "escalate to another model" from "refer to a human" — reported on an honestly stated 322 nodules with patient-clustered confidence intervals. References [7] and [13] each supply half of this; neither supplies both.

Recommended additions to the plan

1. **Enlarge the dataset before claiming anything.** At 322 nodules our AUROC interval is roughly ± 0.03 , wide enough that the Task 5 comparison may not reach significance either way. Every strong result reviewed here rests on a substantially larger cohort.
2. **Add a deep ensemble arm** to Task 6, on the evidence of [7] and [10].
3. **Treat the 2.5/3.5 boundary as a research question**, following [11], rather than a preprocessing footnote.
4. **Use the cached radiologist concepts as a training signal**, not only for explanation — [4] shows the gain, [12] shows the interpretable form.

12 Limitations of this review

This is a targeted review, not a systematic one. There was no PRISMA protocol, no formal inclusion criteria and no assessment of inter-reviewer agreement; searches covered eight themes chosen to match this project's tasks, so coverage is deliberately shaped by our needs rather than by the field's distribution.

Metadata for every entry was verified against the source. Several sources are paywalled — notably [7] and [8] — so those summaries reflect abstract-level and publication-record detail only, and the quantitative claims attributed to them are the ones the authors state in their abstracts. Reference [14] is included for its framework; its author list was not confirmed and must be checked before formal citation.

Numbers quoted across studies are not mutually comparable, for the protocol reasons set out in section 4. They should be read as indicating where the field sits, not as a leaderboard.

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